US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12415250
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Burres, Jr.; Donald Eugene et al.

Clamp for woodworking

Abstract

The present invention provides a clamp for woodworking, The clamp comprises a body having a first end and an opposing second end, and a slot defined on the first end and comprising a plurality of graduated levels extending toward the second end. Each graduated level of the plurality of graduated levels comprises beveled edges.

Inventors: **Burres, Jr.; Donald Eugene (East Palatka, FL), Bernhard, Sr.; Brendan Jude (Palatka, FL)**

Applicant: **Burres, Jr.; Donald Eugene** (East Palatka, FL); **Bernhard, Sr.; Brendan Jude** (Palatka, FL)

Family ID: **1000008508298**

Appl. No.: **19/070023**

Filed: **March 04, 2025**

Publication Classification

Int. Cl.: **B25B5/00** (20060101); **B25B5/14** (20060101)

U.S. Cl.:

CPC **B25B5/145** (20130101);

Field of Classification Search

CPC: B25B (5/145); B25B (5/00); B25B (5/14)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4570345	12/1985	Cattadoris	N/A	N/A
5101932	12/1991	Trudgeon	269/902	E04D 12/008
5280891	12/1993	Estes	269/902	B23Q 3/105
5673898	12/1996	Michalo	29/253	A62B 3/005
6134798	12/1999	Duncan	83/167	G01B 5/0004
8376340	12/2012	Sandmeier	N/A	N/A
9114547	12/2014	Kinsler	N/A	N/A
10435948	12/2018	Moreno	N/A	E06C 7/14

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111633279	12/2019	CN	N/A
1580442	12/2004	EP	N/A

OTHER PUBLICATIONS

Bora, Parallel Clamp Blocks 4-Pack, Accessed Nov. 7, 2024. cited by applicant
Rockler, Rockler Parallel Clamp Blocks, 4-Pack, Accessed Nov. 7, 2024. cited by applicant
Norfolk Wood Shop, 4 pc. Clamp Blocks Set, Accessed Nov. 7, 2024. cited by applicant
Plafope, 4pcs Plywood Accessories Band Clamp Block for Woodworking Band Clamp Block Woodworking Band Clamps Block for Woodworking Ratchet Band Clamp Block Band Clamps Blocks Woodworking, Accessed Nov. 7, 2024. cited by applicant
TDS Online Stores, Multi Functional Anvil-Bench Block with V Slot, Accessed Nov. 7, 2024. cited by applicant

---

Primary Examiner: Wilson; Lee D

Attorney, Agent or Firm: Lippes Mathias LLP

Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure pertains to the field of art of clamps. More specifically, the disclosure pertains to a clamp for use in woodworking that facilitates holding of wood panels while cutting the wood panels with a saw.

PRIOR ART

(2) Wood panels, such as plywood, are typically sold in standard large sheets that need to be resized to fit the specific dimensions an individual requires. Nevertheless, it may be challenging for an individual to cut wood panels solely with a saw operator.

(3) For example, during the process of sawing a wood panel, an unsupported cut portion of the wood panel can deflect downwards relative to a remaining portion of the wood panel. As the cutting of the wood panel progresses, the deflection of the cut portion relative to the remainder of the wood panel can increase. As a result, the cut portion may easily break off and create rough, jagged edges on the wood panel.

(4) Furthermore, a person cutting the wood panel may have a higher risk of encountering hazardous scenarios, such as a saw blade of the saw being bound up, causing a sudden kickback. Currently, to obtain accurate cutting, two people have to be present, one person holding the cut portion and the

remaining portion together, and the other person operating the saw operator. However, this may increase labor costs and expenses for additional personnel.
(5) Therefore, there is a need for a device to assist with woodworking that allows an individual to cut wood panels solely and effortlessly.
SUMMARY OF THE DISCLOSURE
(6) The following is a concise summary of the disclosure presented herein with the primary aim of providing a preliminary understanding of certain aspects of the disclosure. It should be noted, however, that this summary is not intended to serve as a comprehensive overview of the disclosure, nor does it seek to identify or describe any critical or significant elements of the disclosure or the boundaries of its scope. Its sole purpose is to provide a rudimentary understanding of the disclosure's concepts and features, which will be expounded upon in greater detail in the ensuing sections.
(7) In some embodiments, a clamp for woodworking comprises a body having a first end and an opposing second end; and a slot defined on the first end and comprising a plurality of graduated levels extending toward the second end.
(8) In some embodiments, the clamp for woodworking comprises a body having a slot extending partially thereon. The slot comprises a plurality of graduated levels.
(9) The present disclosure pertains to a clamp for use in woodworking designed to securely hold and perfectly align a cut portion and a remaining portion of a wood panel along a kerf during saw cutting, advantageously allowing an individual to cut the wood panel with precision and a smooth finish on edges of the wood panel. The securement of the clamp to the wood panel not only eliminates the dependency on an additional person during cutting of the wood panel, but also mitigates the risk of perilous circumstances, such as an occurrence of a saw's kickback force and handling splinters of the wood panel.
(10) Furthermore, the disclosed clamp features a slot having a plurality of graduated levels that advantageously accommodate and secure wood panels with different thicknesses, thereby providing enhanced versatility and flexibility. Notably, each graduated level of the plurality of graduated levels may have beveled edges to facilitate the wood panel to slide to an appropriate graduated level. In addition, the beveled edges may minimize the risk of injury to a person and the risk of scratching or denting the wood panels, thereby bolstering safety and preventing damage to the wood panels. Moreover, the slot of the disclosed clamp has a symmetrical configuration that allows for simple manufacturing thereof, and provides elevated aesthetic appeal.
(11) The above features and advantages will become apparent from the following detailed description taken with the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** illustrates a front perspective view of a non-limiting exemplary embodiment (the “exemplary embodiment”) of a clamp for woodworking (the “clamp”).
(2) FIG. **2** illustrates a side view with a partially enlarged view of the clamp of FIG. **1**.
(3) FIG. **3** illustrates a first in-use view of the clamp of FIG. **1**.
(4) FIG. **4** illustrates a second in-use view of the clamp of FIG. **1**.
(5) FIG. **5** illustrates a third in-use view of the clamp of FIG. **1**.
NUMBERING REFERENCE
(6) **1**—Clamp **10**—Body **10**A—First Surface **10**B—Second Surface **20**—Slot **21**—Graduated Levels **211**—First graduated level **212**—Second graduated level **213**—Third graduated level **214**—Fourth graduated level **215**—Fifth graduated level **216**—Sixth graduated level **220**—Beveled Edges **101**—Length **102**—Width **103**—Height **301**—Exemplary User **302**—Wood panel **302**_a_—

cut portion **302***b*—remaining portion

DETAILED DESCRIPTION OF THE DISCLOSURE

(7) The following detailed description and accompanying drawings provide a comprehensive disclosure of an exemplary embodiment for the purpose of facilitating one of ordinary skill in the relevant art to make and use the disclosure. As such, the detailed description and illustration of the one or more exemplary embodiments presented herein are purely exemplary in nature and are not intended to limit the scope of the disclosure or its protection in any matter. It is further noted that the drawings may not be to scale, and in some cases, certain details may be omitted which are not necessary for an understanding of the present disclosure, such as conventional details of fabrication and assembly. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description.

(8) For purposes of description herein, the terms “left”, “right”, “front”, “top”, “bottom”, “side” and derivatives thereof shall relate to the device as oriented in FIG. **1**. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description.

(9) Wood panels are versatile and widely used material in construction and crafting. To ensure an accurate cut of the wood panel, and to avoid the need for extra finishes, it is crucial to align a cut portion and a remaining portion of a wood panel during saw cutting while also avoiding potential drooping of the cut portion. The following discloses a clamp for use in woodworking designed to securely hold and perfectly align a cut portion and a remaining portion of a wood panel along a kerf during saw cutting.

(10) A non-limiting, exemplary embodiment (herein after as the “exemplary embodiment”) of a clamp for woodworking **1** (herein after referred to as “the clamp”) is disclosed herein. The exemplary embodiment of the clamp **1** may not only secure the cut portion of the wood panel and the remaining portion of the wood panel but can also be used to accommodate a wide variety of wood panel sizes during saw cutting.

(11) FIG. **1** illustrates a front view of the clamp **1** according to the exemplary embodiment. As shown in FIG. **1**, the clamp **1** comprises a body **10** having a first end comprising a first surface **10**A, an opposing second end comprising a second surface **10**B, and an elongated slot **20** extending from the first surface **10**A towards the second surface **10**B. For example, the slot **20** is defined on the first surface **10**A with a plurality of graduated levels **21** (six of which are shown in FIG. **1**, although the slot **20** may have any suitable numbers of granulated levels **21**). The plurality of granulated levels **21** extends towards the second surface **10**B. As shown in FIG. **1**, the plurality of granulated levels **21** progressively narrow toward the second surface **10**B. The body **10**, as shown in FIG. **1**, has a substantially rectangular prism configuration comprising a length **101**, a width **102**, and a height **103**, with a length-to-width-to-height ratio of 16 to 7 to 4, although any suitable shape and proportions is contemplated. The length **101** of the body **10** may range from 7.99 to 8.01 inches, the width **102** may range from 3.49 to 3.51 inches, and the height **103** may range from to 1.99 to 2.01 inches, although any dimensions is contemplated. The body **10** may be constructed of rigid materials including, but not limited to, plastic, metal and wood.

(12) With continuing reference to FIG. **1**, FIG. **2** illustrates a side view of the clamp **1** of the clamp **1**. As shown in FIG. **2**, the slot **20** is symmetrical along a centroidal axis a-a of the body **10**, although any suitable configuration is contemplated. The plurality of graduated levels **21** include a first graduated level **211**, a second graduated level **212**, a third graduated level **213**, a fourth graduated level **214**, a fifth graduated level **215**, and a sixth graduated level **216** (although any suitable number of graduated levels are contemplated). Each of the granulated levels **211**, **212**, **213**, **214**, **215**, **216** have descending heights from right to left as shown in FIG. **2**. The descending heights between adjacent graduated levels **21** decrease incrementally. For example, the first graduated level **211** can have a first height h**1** of 1.02 to 1.04 inches, the second graduated level

212 can have a second height h**2** of 0.77 to 0.79 inch, the third graduated level **213** can have a third height h**3** of 0.65 to 0.67 inch, the fourth graduated level **214** can have a fourth height h**4** of 0.52 to 0.54 inch, the fifth graduated level **215** can have a fifth height 0.40 to 0.42 inch, and the sixth graduated level **216** can have a sixth height h**6** of 0.27 to 0.29 inch, although any suitable dimensions are contemplated. Moreover, the plurality of graduated levels **21** may have an equal width w of 0.5 inch. The plurality of graduated levels **21** can also have a distance d from the second surface **10**B of the body **10** to the slot **20**, with the distance d being, for example, 0.5 inches. In some embodiments, a select graduated level **21** with a greater height relative to an adjacent graduated level **21** can accommodate a wood panel with a greater thickness, whereas a select graduated level **21** with a shorter height can hold a wood panel with a reduced thickness.

(13) With continuing reference to FIGS. **1** and **2**, each graduated level of the plurality of graduated levels **21** comprises beveled edges **220** having a predetermined angle a**1** relative to an adjoining surface on which the clamp **1** rests. The predetermined angle a**1** of the beveled edges **220** is preferably 45 degrees, as shown in FIG. **2**, or may range from 30 degrees to 60 degrees, although any suitable angle is contemplated. Furthermore, each of the beveled edges **220** may have a length of 0.0615 to 0.0635 inch. It is also anticipated that the beveled edges **220** may be curved or round edges to enhance safety and prevent damage of the wood panels.

(14) Referring to FIGS. **3** to **5**, an exemplary method of using the clamp **1** in woodworking is presented herein to further demonstrate the convenience and advantages of the clamp **1**. It is anticipated that several steps may be sequentially interchangeable and equivalent application of one or more permutations of such sequentially interchangeable steps does not alter the spirit of the disclosure in any meaningful way.

(15) As shown in FIG. **3**, a wood panel **302**, such as plywood, is positioned on a worktable or sawhorse **303**, and an exemplary user **301** initiates the saw cutting process by sawing a part of the wood panel **302** and dividing the wood panel **302** into a cut portion **302***a* and a remaining portion **302***b*. During a typical saw cutting processes, the cut portion **302***a* tends to droop downwards relative to the remaining portion **302***b* of the wood panel **302**, as shown in FIG. **3**.

(16) Next, as shown in FIG. **4**, the exemplary user **301** may effortlessly place and move the slot **20** of the clamp **1** towards the cut and remaining portions **302***a*, **302***b* to eliminate the droop. To do so, the exemplary user **301** positions the clamp **10** adjacent to the cut and remaining portions **302***a*, **302***b*, and moves the clamp **10** towards the cut and remaining portions **302***a*, **302***b* until the cut and remaining portions **302***a*, **302***b* engage at least one of the beveled edges **220** and continues moving until engaging one of graduated levels **21** in the slot **20** having a width equal to that of the cut and remaining portions **302***a*, **302***b* with a friction fit. The clamp **10** then engages the cut and remaining portions **302***a*, **302***b* with the friction fit to secure the clamp **10** to the cut and remaining portions **302***a*, **302***b*. The beveled edges **220** of the engaged graduated level **21**, engaged with the cut portion **302***a* and the remaining portion **302***b*, to become aligned and stationary relative to each other. As a result, the appropriate graduated level **21** holds the cut portion **302***a* and the remaining portion **302***b* together to align the cut portion **302***a* and the remaining portion **302***b*, advantageously substantially maintaining the cut line for the exemplary user **301** while also preventing kickback of the saw. The exemplary user **301** may then continue the saw cutting process.

(17) As shown in FIG. **5**, with the clamp **1** holding the cut and remaining portions **302***a*, **302***b* in place, the exemplary user **301** can make a straight and clean kerf for the wood panel **302** effortlessly.

(18) The disclosed clamp **1** not only offers a convenient and safe solution for wood working, but also advantageously provides versatility for accommodating wood panels with a wide range of thicknesses.

(19) While the embodiments of the disclosure have been disclosed, certain modifications may be made by those skilled in the art to modify the disclosure without departing from the spirit of the disclosure.

Claims

1. A clamp for woodworking, the clamp comprising: a body having a first end and an opposing second end; and a slot defined on the first end and comprising a plurality of graduated levels extending toward the second end; wherein the plurality of graduated levels comprises a first graduated level having a height of 1.02 to 1.04 inches, a second graduated level having a height of 0.77 to 0.79 inch, a third graduated level having a height of 0.65 to 0.67 inch, a fourth graduated level having a height of 0.52 to 0.54 inch, a fifth graduated level having a height of 0.40 to 0.42 inch, and a sixth graduated level having a height of 0.27 to 0.29 inch.
2. The clamp of claim 1, wherein the body has a substantially rectangular prism configuration.
3. The clamp of claim 2, wherein the rectangular prism configuration has a length-to-width-to height ratio of 16 to 7 to 4.
4. The clamp of claim 2, wherein the body has a length 7.99 to 8.01 inches; a width of 3.49 to 3.51 inches; and a height of 1.99 to 2.01 inches.
5. The clamp of claim 1, wherein the slot is symmetrical along a centroidal axis of the body.
6. The clamp of claim 1, wherein each graduated level of the plurality of graduated levels has an equal width.
7. The clamp of claim 6, wherein the equal width is 0.5 inch.
8. The clamp of claim 1, wherein each graduated level of the plurality of graduated levels comprises beveled edges.
9. The clamp of claim 8, wherein each of the beveled edges forms a 45-degree angle relative to an associated adjoining surface on which the body rests.
10. The clamp of claim 8, wherein a length of each of the beveled edges is 0.0615 to 0.0635 inch.
11. The clamp of claim 1, wherein a distance from the second end to the slot is 0.5 inch.
12. A clamp for woodworking, the clamp comprising: a body having a slot extending partially thereon; and wherein the slot comprises a plurality of graduated levels; wherein the plurality of graduated levels comprises a first graduated level having a height of 1.02 to 1.04 inches, a second graduated level having a height of 0.77 to 0.79 inch, a third graduated level having a height of 0.65 to 0.67 inch, a fourth graduated level having a height of 0.52 to 0.54 inch, a fifth graduated level having a height of 0.40 to 0.42 inch, and a sixth graduated level having a height of 0.27 to 0.29 inch.
13. The clamp of claim 12, wherein each graduated level of the plurality of graduated levels comprises beveled edges.
14. The clamp of claim 13, wherein each of the beveled edges forms a 45-degree angle relative to an associated adjoining surface on which the body rests.
15. The clamp of claim 14, wherein the slot is symmetrical along a centroidal axis of the body.
16. The clamp of claim 12, wherein the body has a substantially rectangular prism configuration.
17. The clamp of claim 16, wherein the rectangular prism configuration has a length-to-width-to height ratio of 16 to 7 to 4.
18. The clamp of claim 12, wherein each graduated level of the plurality of graduated levels has an equal width of 0.5 inch.
19. A clamp for woodworking, the clamp comprising: a body having a first end and an opposing second end; and wherein the slot comprises a plurality of graduated levels; wherein the body has a substantially rectangular prism configuration having a length 7.99 to 8.01 inches; a width of 3.49 to 3.51 inches; and a height of 1.99 to 2.01 inches; and a slot defined on the first end and comprising a plurality of graduated levels extending toward the second end.
20. The clamp of claim 19, wherein the plurality of graduated levels comprises six graduated levels.